CUBIC-FIT: A High Performance and TCP CUBIC Friendly Congestion Control Algorithm

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Reference

- **Paper Title:** CUBIC-FIT: A High Performance and TCP CUBIC Friendly Congestion Control Algorithm
- **Author:** Jingyuan Wang, Jiangtao Wen, Yuxing Han, Jun Zhang, Chao Li and Zhang Xiong
- **Source:** `https://www.researchgate.net/publication/260533734_CUBIC-FIT_A_high_performance_and_tcp_CUBIC_friendly_congestion_control_algorithm`

Selected Base Paper

Title: CUBIC-FIT: A High Performance and TCP CUBIC Friendly Congestion Control Algorithm

- **Core Problem:** Standard CUBIC treats *all* packet losses as congestion, causing performance drops in wireless networks where random loss occurs.

Plan 1: Dynamic Alpha (α) Adaptation

Proposed Modification: Implement a **Dynamic Gradient Alpha** that adjusts α based on the *rate of change* of RTT

Plan 2: Bottleneck Queue Estimation

Proposed Modification: Implement Smart Virtual Scaling.

Plan 3: Simulated Multi-Flow Aggregation

Proposed Modification: Models the aggression of N parallel flows within a single TCP connection to maximize bandwidth use.